



Urea-Formaldehyde Resin Systems Modified with Partially Pyrolyzed and Delignified Lignin: A Review

Yıl 2025, Cilt: 3 Sayı: 2, 158 - 168, 28.09.2025

Hande Yılmaz*, Doğu Ramazanoğlu, Halil Uğraş, Ayhan Tozluoğlu

<https://doi.org/10.61150/ijonfest.2025030206>

<https://izlik.org/JA89HG33YJ>

Öz

Urea-formaldehyde UF resins are typical adhesives employed in the production of wood-derived products. Application of such fossil fuel-based resins that emit formaldehyde emissions endangers the environmental and human health conditions profoundly. Bio-based lignin polymer can be utilized as a replacement additive in UF resins because it has inherent adhesive and water repellent nature. When lignin is added into the structure of the resin, the C-N bonds are reduced through the creation of new bonds within regions containing methylene bonds. Raw lignin cannot be directly added to UF resins; it must undergo pretreatment due to its poor binding capacity and amorphous nature. Therefore, partial pyrolysis of lignin in the low temperature zone under inert atmosphere was suggested in this study. This reaction creates a longer chain structure by partial C-O bond fragmentation; thus lignin is altered. This method is likely to reduce the formaldehyde emissions, but the same chain structure will increase the water repellency and reduce the moisture content of lignin. Literature research reveals that lignin additives have great effects on resin viscosity, free formaldehyde content, adhesion quality and panel performance.

Anahtar Kelimeler

[lignin](#), [UF Resin](#), [Depolymerization](#), [Partial Pyrolysis](#), [Adhesives](#)

Etik Beyan

This article has been supported by the TÜBİTAK University-Industry Collaboration Support Program through Project No. 5230122 at Istanbul Gedik University.

Destekleyen Kurum

TÜBİTAK, İstanbul Gedik Üniversitesi

Proje Numarası

5230122

Kaynakça

- [1] M. Hassegawa, J. Van Brusselen, M. Cramm, and P. J. Verkerk, "Wood-Based Products in the Circular Bioeconomy: Status and Opportunities towards Environmental Sustainability," *Land (Basel)*, vol. 11, no. 12, p. 2131, Dec. 2022, doi: 10.3390/LAND11122131/S1.
- [2] Á. Galán-Martín, V. Tulus, I. Díaz, C. Pozo, J. Pérez-Ramírez, and G. Guillén-Gosálbez, "Sustainability footprints of a renewable carbon transition for the petrochemical sector within planetary boundaries," *One Earth*, vol. 4, no. 4, pp. 565-583, Apr. 2021, doi: 10.1016/J.ONEEAR.2021.04.001/ATTACHMENT/D5F8717E-7F87-4C2E-B364-370B57DB7741/MMC2.PDF.
- [3] C. Yu, Y. Chen, R. Li, J. Jiang, and X. Wang, "A Narrative Review: Modification of Bio-Based Wood Adhesive for Performance Improvement," *Coatings* 2024, Vol. 14, Page 1153, vol. 14, no. 9, p. 1153, Sep. 2024, doi: 10.3390/COATINGS14091153.
- [4] E. A. R. Zuiderveen et al., "The potential of emerging bio-based products to reduce environmental impacts," *Nat Commun*, vol. 14, no. 1, pp. 1-7, Dec. 2023, doi:10.1038/S41467-023-43797-9;SUBJMET=106,172,2739,4081,682,694,704;KWRD=CLIMATE-CHANGE+IMPACTS,CLIMATE-CHANGE+MITIGATION

N,ENVIRONMENTAL+IMPACT.

- “ [5] S. Ghahri, L. Yang, G. Du, and B. D. Park, “Transition from Formaldehyde-Based Wood Adhesives to Bio-Based Alternatives,” *Bioresources*, vol. 20, no. 2, pp. 2476–2479, 2025, doi: 10.15376/BIORES.20.2.2476–2479.
- “ [6] G. Yang, Z. Gong, X. Luo, L. Chen, and L. Shuai, “Bonding wood with uncondensed lignins as adhesives,” *Nature*, vol. 621, no. 7979, pp. 511–515, Sep. 2023, doi: 10.1038/S41586-023-06507-5;SUBJMETA=224,303,455,638,639,685;KWRD=MECHANICAL+PROPERTIES,SUSTAINABILITY.
- “ [7] W. Tian et al., “Development of High Strength, High Water Resistant, Low-Formaldehyde Emission Urea-Formaldehyde Adhesive and Its Effect on the Properties of Wood Composites,” *J Appl Polym Sci*, vol. 142, no. 10, p. e56548, Mar. 2025, doi: 10.1002/APP.56548;WGROU:STRING:PUBLICATION.
- “ [8] K. Saito, Y. Hirabayashi, and S. Yamanaka, “Reduction of formaldehyde emission from urea-formaldehyde resin with a small quantity of graphe ne oxide,” *RSC Adv*, vol. 11, no. 52, pp. 32830–32836, Oct. 2021, doi: 10.1039/D1RA06717F.
- “ [9] M. A. Aristri et al., “Recent Developments in Lignin- and Tannin-Based Non-Isocyanate Polyurethane Resins for Wood Adhesives—A Review,” *Applied Sciences* 2021, Vol. 11, Page 4242, vol. 11, no. 9, p. 4242, May 2021, doi: 10.3390/AP11094242.
- “ [10] P. Antov, V. Savov, and N. Neykov, “WOOD RESEARCH 65 (1): 2020 xx-xx SUSTAINABLE BIO-BASED ADHESIVES FOR ECO-FRIENDLY WOOD COMPOSITES. A REVIEW”.

Tüm Kaynakçayı Göster ▼

Toplam **66** adet kaynakça vardır.

Ayrıntılar

Birincil Dil	İngilizce
Konular	Orman Endüstri Mühendisliği (Diğer)
Bölüm	Derleme
Yazarlar	Hande Yılmaz* İstanbul Gedik Üniversitesi 0009-0004-9577-323X Türkiye Doğu Ramazanoğlu İSTANBUL GEDİK ÜNİVERSİTESİ, MÜHENDİSLİK FAKÜLTESİ 0000-0002-6356-5792 Türkiye Halil Uğraş BEYKENT ÜNİVERSİTESİ, MÜHENDİSLİK-MİMARLIK FAKÜLTESİ Türkiye Ayhan Tozluoğlu DÜZCE ÜNİVERSİTESİ, ORMAN FAKÜLTESİ, ORMAN ENDÜSTRİSİ MÜHENDİSLİĞİ BÖLÜMÜ Türkiye
Proje Numarası	5230122
Gönderilme Tarihi	8 Temmuz 2025
Kabul Tarihi	16 Temmuz 2025
Yayımlanma Tarihi	28 Eylül 2025
DOI	https://doi.org/10.61150/ijonfest.2025030206
IZ	https://izlik.org/JA89HG33YJ
Yayımlandığı Sayı	Yıl 2025 Cilt: 3 Sayı: 2

Kaynak Göster

IEEE

[1]H. Yılmaz, D. Ramazanoğlu, H. Uğraş, ve A. Tozluoğlu, “Urea-Formaldehyde Resin Systems Modified with Partially Pyrolyzed and Delignified Lignin: A Review”, *IJONFEST*, c. 3, sy 2, ss. 158–168, Eyl. 2025, doi: 10.61150/ijonfest.2025030206.



• **MAKALE DOSYALARI**

– [Tam Metin](#)



👁 114

↓ 711

☆ 0

• [Dergi Ana Sayfası](#)

• [Arşiv](#)

